# The Parallel Multiple Mediator Model with Multiple Causal Antecedents and Consequent Variables

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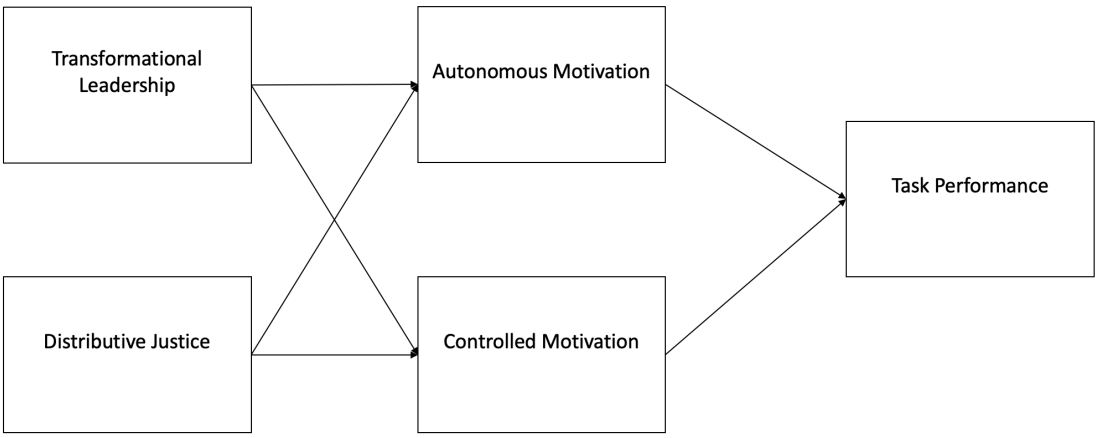
## Problem

Why is it that some people have higher task performance and lower job burnout whereas others don't?

## Theory

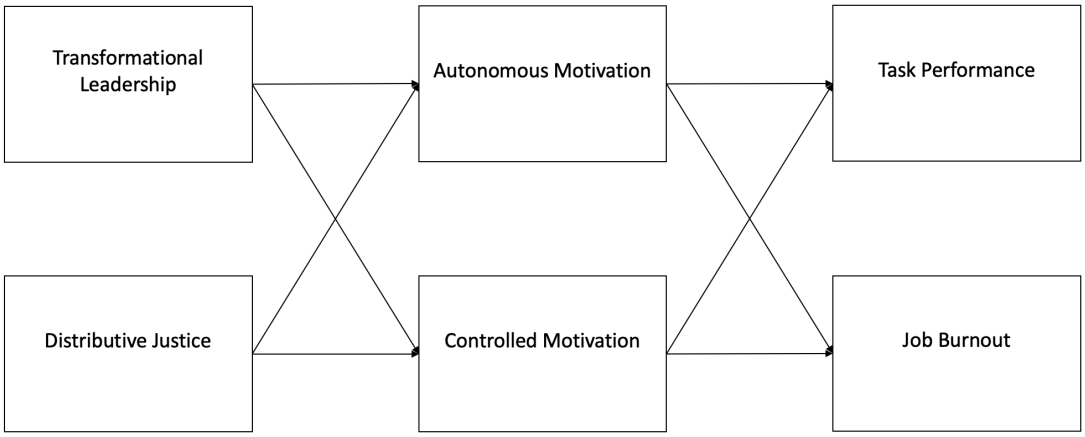
According to self-determination theory, individuals' work behaviors are predicted by their motivations (autonomous and controlled), which, in turn, are predicted by workplace context (e.g., leadership style and distributive justice).




We want to test if this model is true or not. So, we go out to collect some data but what should we do next?

## Notation: Diagram

- **X1**: causal antecedent variable1
- **X2**: causal antecedent variable2
- **Y1**: consequent variable 1
- **Y2**: consequent variable 2
- **M1** "mediator 1"
- **M2** "mediator 2"



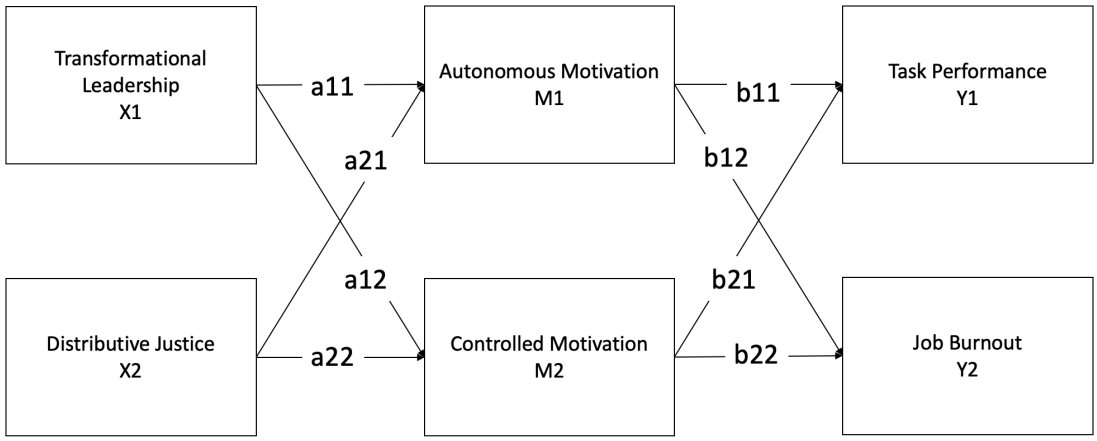

## Activity

- Use the notation shown here to draw the relation you would like to investigate.

## Notation: Indirect effects

- Let a11, a12, a21, a22, b11, b12, b21, and b22 be quantifications of causal effects, such as regression coefficients in an ordinary least squares model (or maximum likelihood path estimates in a structural equation model).
- **a11 × b11 = "indirect effect" of X1 on Y1 (This is our main focus.)**
- **a21 × b11 = "indirect effect" of X2 on Y1 (This is our main focus.)**
- **a12 × b21 = "indirect effect" of X1 on Y1 (This is our main focus.)**
- **a22 × b21 = "indirect effect" of X2 on Y1 (This is our main focus.)**
- **a11 × b12 = "indirect effect" of X1 on Y2 (This is our main focus.)**
- **a21 × b12 = "indirect effect" of X2 on Y2 (This is our main focus.)**
- **a12 × b22 = "indirect effect" of X1 on Y2 (This is our main focus.)**
- **a22 × b22 = "indirect effect" of X2 on Y2 (This is our main focus.)**

## Regression Equation




$$AutonomousMotivation_j = i + a11TransformationalLeadership_j + a21DistributiveJustice_j + \varepsilon_j$$

$$ControlledMotivation_j = i + a12TransformationalLeadership_j + a22DistributiveJustice_j + \varepsilon_j$$

$$TaskPerferomance_j = i + b11AutonomousMotivation_j + b21ControlledMotivation_j + \varepsilon_j$$

$$JobBurnout_j = i + b12AutonomousMotivation_j + b22ControlledMotivation_j + \varepsilon_j$$

## Transformational Leadership

Transformational leadership is a leadership style that can inspire positive changes in those who follow. Transformational leaders are generally energetic, enthusiastic, and passionate. Not only are these leaders concerned and involved in the process, but they are also focused on helping every member of the group succeed.

## Distributive Justice

Distributive justice refers to the ethical principle concerned with the fair allocation of resources, benefits, and burdens among members of a society or group. It focuses on how goods, wealth, opportunities, and responsibilities should be distributed to ensure fairness, equity, and the well-being of individuals.

## Job Buronout

Job burnout is a psychological syndrome characterized by prolonged exposure to chronic work-related stress, resulting in emotional exhaustion, depersonalization (cynicism or detachment from the job), and a reduced sense of personal accomplishment. It is often associated with overwhelming demands, insufficient resources, and a lack of balance between effort and reward in the workplace.

## Run Process Macro

Whenever you would like to run conditional process analysis, don't forget to open and run the script of Process Macro!

```
******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

PROCESS is now ready for use.
Copyright 2020-2023 by Andrew F. Hayes ALL RIGHTS RESERVED
Workshop schedule at http://haskayne.ucalgary.ca/CCRAM
```

## Load Packages and Import Data

```
library(tidyverse)
library(psych)
library(haven)

data <- read_sav("2x2m2y.sav")

#view our data
data

# A tibble: 500 × 6
   leaders justice  auto    cm performance  burn
     <dbl>   <dbl> <dbl> <dbl>       <dbl> <dbl>
 1    3.75    3.21  1.33  2.82        36.3  3.38
 2    4.59    4.47  3.79  2.38        58.3  2.37
 3    4.45    3.56  3.21  1.16        38.5  1.08
 4    3.21    3.95  3.27  1.06        39.1  1.42
 5    3.79    3.07  4.51  1.45        36.0  1.97
 6    3.77    4.07  4.47  3.20        45.2  3.19
 7    2.52    3.30  3.18  2.92        19.0  3.29
 8    3.79    2.74  6.00  1.85        43.4  2.56
 9    1.73    2.65  3.42  2.48        15.0  3.08
10    5.15    3.27  3.29  2.36        51.9  3.53
# ℹ 490 more rows

#descriptive statistics
describe(data)

            vars   n  mean    sd median trimmed   mad  min    max  range  ske
w
leaders        1 500  4.00  0.97   4.02    4.02  0.98 1.00   6.79   5.79 -0.1
5
```

```
justice        2 500  4.03  0.96   4.01    4.03  1.00 1.00   6.75   5.75  0.0
0
auto           3 500  4.05  0.99   4.06    4.05  0.99 1.31   7.00   5.69  0.0
0
cm             4 500  2.84  1.00   2.83    2.84  1.06 1.00   6.13   5.13  0.1
4
performance    5 500 44.83 14.66  44.20   44.87 15.42 0.00 100.00 100.00  0.0
7
burn           6 500  2.82  1.03   2.77    2.80  1.09 1.00   6.71   5.71  0.2
6
            kurtosis   se
leaders         0.01 0.04
justice        -0.32 0.04
auto            0.12 0.04
cm             -0.34 0.04
performance     0.07 0.66
burn           -0.20 0.05
```

### Visualize (Bivariate Correlations)

```
data |> as.matrix() |> Hmisc::rcorr()
```

```
            leaders justice  auto    cm performance  burn
leaders        1.00    0.57  0.49 -0.27        0.38 -0.26
justice        0.57    1.00  0.52 -0.17        0.38 -0.12
auto           0.49    0.52  1.00 -0.06        0.27 -0.06
cm            -0.27   -0.17 -0.06  1.00       -0.07  0.58
performance    0.38    0.38  0.27 -0.07        1.00 -0.08
burn          -0.26   -0.12 -0.06  0.58       -0.08  1.00

n= 500



P
            leaders justice auto   cm     performance burn
leaders             0.0000  0.0000 0.0000 0.0000      0.0000
justice     0.0000          0.0000 0.0000 0.0000      0.0054
auto        0.0000  0.0000         0.2116 0.0000      0.1478
cm          0.0000  0.0000  0.2116        0.1031      0.0000
performance 0.0000  0.0000  0.0000 0.1031              0.0766
burn        0.0000  0.0054  0.1478 0.0000 0.0766
```

```
#data |> select(leader, justice, am, cm, performance) |> as.matrix() |> Hmisc
::rcorr()
```

### Model (conditional process analysis: the parallel multiple mediator model with multiple causal antecedents and multiple consequnt variables)

Unfortunately, there is no existing model we can choose to run the parallel multiple mediator analysis with multiple causal antecedents. However, the conditional process

analysis framework allows us to create and customize our own model using bmatrix and cmatrix. This is how:

| | X1 | M1 | M2 |
|---|---|---|---|
| M1 | 1 | | |
| M2 | 1 | 0 | |
| Y1 | 0 | 1 | 1 |
| (1,1,0,0,1,1) | | | |
| | | | |
| | X1 | M1 | M2 |
| M1 | 1 | | |
| M2 | 1 | 0 | |
| Y2 | 0 | 1 | 1 |
| (1,1,0,0,1,1) | | | |

| | c1 |
|---|---|
| M1 | 1 |
| M2 | 1 |
| Y1 | 0 |
| | |
| (1,1,0) | |
| | |

| | c1 |
|---|---|
| M1 | 1 |
| M2 | 1 |
| Y2 | 0 |
| | |
| (1,1,0) | |

```
process(data = data, y = "performance", x = "leaders", cov = c("justice"), m
= c("auto", "cm"), seed = 12345, bmatrix = c(1, 1, 0, 0, 1, 1), cmatrix = c(1
,1,0))


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************
```

```
Model : CUSTOM
    Y : performance
    X : leaders
   M1 : auto
   M2 : cm

Covariates:
       justice

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: auto

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5687     0.3234     0.6701   118.7716     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.4054     0.1758     7.9943     0.0000     1.0600     1.7507
leaders      0.2939     0.0459     6.3957     0.0000     0.2036     0.3841
justice      0.3654     0.0463     7.8990     0.0000     0.2745     0.4563

**************************************************************************
Outcome Variable: cm

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2725     0.0743     0.9235    19.9338     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     4.0108     0.2064    19.4339     0.0000     3.6054     4.4163
leaders     -0.2629     0.0539    -4.8740     0.0000    -0.3689    -0.1569
justice     -0.0288     0.0543    -0.5307     0.5959    -0.1355     0.0779

**************************************************************************
Outcome Variable: performance

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2723     0.0742   199.7120    19.9041     2.0000   497.0000     0.0000

Model:
```

```
             coeff        se         t         p      LLCI      ULCI
constant   31.5492    3.2976    9.5673    0.0000   25.0702   38.0282
auto        3.8781    0.6380    6.0786    0.0000    2.6246    5.1316
cm         -0.8569    0.6357   -1.3480    0.1783   -2.1058    0.3921

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Number of bootstraps for percentile bootstrap confidence intervals: 5000

process(data = data, y = "performance", x = "justice", cov = c("leaders"), m
= c("auto", "cm"), seed = 12345, bmatrix = c(1, 1, 0, 0, 1, 1), cmatrix = c(1
,1,0))


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : CUSTOM
    Y : performance
    X : justice
   M1 : auto
   M2 : cm

Covariates:
       leaders

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: auto

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5687     0.3234     0.6701   118.7716     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.4054     0.1758     7.9943     0.0000     1.0600     1.7507
justice      0.3654     0.0463     7.8990     0.0000     0.2745     0.4563
leaders      0.2939     0.0459     6.3957     0.0000     0.2036     0.3841

**************************************************************************
Outcome Variable: cm

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2725     0.0743     0.9235    19.9338     2.0000   497.0000     0.0000
```

```
Model:
             coeff         se          t          p       LLCI       ULCI
constant    4.0108     0.2064    19.4339     0.0000     3.6054     4.4163
justice    -0.0288     0.0543    -0.5307     0.5959    -0.1355     0.0779
leaders    -0.2629     0.0539    -4.8740     0.0000    -0.3689    -0.1569

**********************************************************************
Outcome Variable: performance

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2723     0.0742   199.7120    19.9041     2.0000   497.0000     0.0000

Model:
             coeff         se          t          p       LLCI       ULCI
constant   31.5492     3.2976     9.5673     0.0000    25.0702    38.0282
auto        3.8781     0.6380     6.0786     0.0000     2.6246     5.1316
cm         -0.8569     0.6357    -1.3480     0.1783    -2.1058     0.3921

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```
**************** DIRECT AND INDIRECT EFFECTS OF X ON Y ****************

The direct effect of X on Y is fixed to zero.

Indirect effect(s) of X on Y:
         Effect    BootSE  BootLLCI  BootULCI
TOTAL    1.4417    0.3079    0.8432    2.0663
auto     1.4170    0.3074    0.8283    2.0323
cm       0.0247    0.0572   -0.0743    0.1670

******************* ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output: 95

Number of bootstraps for percentile bootstrap confidence intervals: 5000
```

```
process(data = data, y = "burn", x = "leaders", cov = c("justice"), m = c("au
to", "cm"), seed = 12345, bmatrix = c(1, 1, 0, 0, 1, 1), cmatrix = c(1,1,0))
```

```
******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : CUSTOM
    Y : burn
    X : leaders
   M1 : auto
   M2 : cm

Covariates:
       justice

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: auto

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5687     0.3234     0.6701   118.7716     2.0000   497.0000     0.0000

Model:
```

```
             coeff        se         t         p      LLCI      ULCI
constant    1.4054    0.1758    7.9943    0.0000    1.0600    1.7507
leaders     0.2939    0.0459    6.3957    0.0000    0.2036    0.3841
justice     0.3654    0.0463    7.8990    0.0000    0.2745    0.4563

**********************************************************************
Outcome Variable: cm

Model Summary:
          R      R-sq       MSE         F       df1       df2         p
     0.2725    0.0743    0.9235   19.9338    2.0000  497.0000    0.0000

Model:
             coeff        se         t         p      LLCI      ULCI
constant    4.0108    0.2064   19.4339    0.0000    3.6054    4.4163
leaders    -0.2629    0.0539   -4.8740    0.0000   -0.3689   -0.1569
justice    -0.0288    0.0543   -0.5307    0.5959   -0.1355    0.0779

**********************************************************************
Outcome Variable: burn

Model Summary:
          R      R-sq       MSE         F       df1       df2         p
     0.5821    0.3388    0.6976  127.3428    2.0000  497.0000    0.0000

Model:
             coeff        se         t         p      LLCI      ULCI
constant    1.2594    0.1949    6.4618    0.0000    0.8765    1.6423
auto       -0.0334    0.0377   -0.8867    0.3757   -0.1075    0.0407
cm          0.5958    0.0376   15.8596    0.0000    0.5220    0.6697

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Outcome Variable: auto

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5687     0.3234     0.6701   118.7716     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.4054     0.1758     7.9943     0.0000     1.0600     1.7507
justice      0.3654     0.0463     7.8990     0.0000     0.2745     0.4563
leaders      0.2939     0.0459     6.3957     0.0000     0.2036     0.3841

**********************************************************************
Outcome Variable: cm

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2725     0.0743     0.9235    19.9338     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     4.0108     0.2064    19.4339     0.0000     3.6054     4.4163
justice     -0.0288     0.0543    -0.5307     0.5959    -0.1355     0.0779
leaders     -0.2629     0.0539    -4.8740     0.0000    -0.3689    -0.1569

**********************************************************************
Outcome Variable: burn

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5821     0.3388     0.6976   127.3428     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.2594     0.1949     6.4618     0.0000     0.8765     1.6423
auto        -0.0334     0.0377    -0.8867     0.3757    -0.1075     0.0407
cm           0.5958     0.0376    15.8596     0.0000     0.5220     0.6697

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**************** DIRECT AND INDIRECT EFFECTS OF X ON Y ****************

The direct effect of X on Y is fixed to zero.

Indirect effect(s) of X on Y:
         Effect    BootSE  BootLLCI  BootULCI
TOTAL   -0.0294    0.0352   -0.0986    0.0411
auto    -0.0122    0.0134   -0.0385    0.0149
cm      -0.0172    0.0310   -0.0772    0.0453

******************* ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output: 95

Number of bootstraps for percentile bootstrap confidence intervals: 5000

#data = the name of your dataset
#y = consequent variable
#x = causal antecedent variable
#m = mediator
#seed = for the purpose of replication
#bmatrix = specify how x1, m, and y should be related
#cmatrix = specify how x2, m, and y should be related

#evaluation: overall model + predictor(s)
```

```
#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the pr
edictability of our model. The higher the better.

#Outcome Variable: auto => path a11 and path a21

#interpretation: About 32.34% of variance in autonomous motivation is explain
ed by (transformational) leadership style and distributive justice

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and positively predicts autonomous
motivation. One unit increase in leadership is expected to be associated with
 an increase of 0.2939 units in autonomous motivation.

#interpretation: Distributive justice significantly and positively predicts a
utonomous motivation. One unit increase in leadership is expected to be assoc
iated with an increase of0.3654 units in autonomous motivation.

#Outcome Variable: control => path a12 and path a22

#interpretation: About 7.43% of variance in controlled motivation is explaine
d by (transformational) leadership style and distributive justice.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and negatively predicts controlled
motivation. One unit increase in leadership is expected to be associated with
 a decrease of 0.2629 units in controlled motivation.

#interpretation: Distributive justice does not significantly predict controll
ed motivation. One unit increase in distributive justice is expected to be as
sociated with a non-significant decrease of 0.0288 units in controlled motiva
tion.

#Outcome Variable: performance => path b11 and path b21

#interpretation: About 7.42% of variance in task performance is explained by
autonomous motivation AND controlled motivation.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)
```

```
#interpretation b11: Autonomous motivation significantly and positively predi
cts task performance. One unit increase in autonomous motivation is expected
to be associated with an increase of 3.8781 units in task performance.

#interpretation b21: Controlled motivation does not significantly predict tas
k performance. One unit increase in controlled motivation is expected to be a
ssociated with an non-significant decrease of 0.8569 units in task performanc
e.

#Outcome Variable: performance => path b12 and path b22

#interpretation: About 33.88% of variance in job burnout is explained by auto
nomous motivation AND controlled motivation.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b12: Autonomous motivation does not significantly predicts jo
b burnout. One unit increase in autonomous motivation is expected to be assoc
iated with a non-significant decrease of 0.0334 units in job burnout.

#interpretation b22: Controlled motivation significantly and positively predi
ct job burnout. One unit increase in controlled motivation is expected to be
associated with an increase of 0.5958  units in job burnout.

#INDIRECT EFFECTS OF X1 ON Y1
#Indirect effect(s) of X on Y => a11 x b11 AND a12 x b21
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a11 x b11: Autonomous motivation significantly mediates the r
elation between leadership and task performance (a11b11 = 1.1396, 95% CI [0.6
580, 1.7003]).

#interpretation a11 x b11: Leadership is indirectly related to task performan
ce through the pathway of autonomous motivation (a11b11 = 1.1396, 95% CI [0.6
580, 1.7003]).

#interpretation a12 x b21: Controlled motivation does not significantly media
te the relation between leadership and task performance (a12b21 = 0.2253, 95%
 CI [-0.0706, 0.6021]).

#interpretation a12 x b21: Leadership is not indirectly related to task perfo
rmance through the pathway of controlled motivation (a12b21 = 0.2253, 95% CI
[-0.0706, 0.6021]).
```

```
#INDIRECT EFFECTS OF X2 ON Y1
#Indirect effect(s) of X on Y => a21 x b11 AND a22 x b21
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a21 x b11: Autonomous motivation significantly mediates the r
elation between distributive justice and task performance (a21b11 = 1.4170, 9
5% CI [0.8283, 2.0323]).

#interpretation a21 x b11: Distributive justice is indirectly related to task
 performance through the pathway of autonomous motivation (a21b11 = 1.4170, 9
5% CI [0.8283, 2.0323]).

#interpretation a22 x b21: Controlled motivation does not significantly media
te the relation between distributive justice and task performance (a22b21 = 0
.0247, 95% CI [-0.0743, 0.1670).

#interpretation a22 x b21: Distributive justice is not indirectly related to
task performance through the pathway of controlled motivation (a22b21 = 0.024
7, 95% CI [-0.0743, 0.1670]).

#INDIRECT EFFECTS OF X1 ON Y2
#Indirect effect(s) of X on Y => a11 x b12 AND a12 x b22
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a11 x b12: Autonomous motivation does not significantly media
te the relation between leadership and job burnout (a11b12 = -0.0098, 95% CI
[-0.0329, 0.0107]).

#interpretation a11 x b12: Leadership is not indirectly related to job burnou
t through the pathway of autonomous motivation (a11b12 = -0.0098, 95% CI [-0.
0329, 0.0107]).

#interpretation a12 x b22: Controlled motivation significantly mediates the r
elation between leadership and job burnout (a12b22 = -0.1567, 95% CI [-0.2262
, -0.0907]).

#interpretation a12 x b22: Leadership is indirectly related to job burnout th
rough the pathway of controlled motivation (a12b22 = -0.1567, 95% CI [-0.2262
, -0.0907]).
```

```
#INDIRECT EFFECTS OF X2 ON Y2
#Indirect effect(s) of X on Y => a21 x b12 AND a22 x b22
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a21 x b11: Autonomous motivation does not significantly media
te the relation between distributive justice and job burnout (a21b12 = -0.012
2, 95% CI [-0.0385, 0.0149]).

#interpretation a21 x b11: Distributive justice is not indirectly related to
job burnout through the pathway of autonomous motivation (a21b12 = -0.0122, 9
5% CI [-0.0385, 0.0149]).

#interpretation a22 x b21: Controlled motivation does not significantly media
te the relation between distributive justice and job burnout (a22b22 = -0.017
2, 95% CI [-0.0772, 0.0453]).

#interpretation a22 x b21: Distributive justice is not  indirectly related to
 job burnout through the pathway of controlled motivation (a22b22 = -0.0172,
95% CI [-0.0772, 0.0453]).

#Conclusion: When employees perceived their supervisors to be more transforma
tional and rewards are distributed more fairly, the higher autonomous motivat
ion they have, which, in turn, is significantly and positively associated wit
h task performance. Furthermore, when employees perceived their supervisors t
o be more transformational, the lower controlled motivation they have, which,
 in turn, is significantly associated with lower job burnout.
```

## Assignment 5-1

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) and the income level of child's household (**povlev4_21**) serve as causal antecedent variables, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) and weight status (**BMI4_21**) serves as consequent variables. Please run a conditional process analysis to find out the answer. (Let's assume that these six variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 5_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - income level of child's household (**povlev4_21**) : Low values indicate better lower income.

- sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.
- fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
- vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
- teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
- weight status (**BMI4_21**): High values indicate overweight.

- Draw the diagram to represent the paralell multiple mediator model with multiple causal antecedents and consequent variables.
- Write down the bmatrix.
- Write down the cmatrix.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a11, a12, a13, a21, a22, a23, b11, b21, b31, b12, b22, b32, a11 x b11, a12 x b21, a13 x b31, a21 x b11, a22 x b21, a23 x b31, a11 x b12, a12 x b22, a13 x b32, a21 x b12, a22 x b22, and a23 x b32
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a11, a12, a13, a21, a22, a23, b11, b21, b31, b12, b22, b32, a11 x b11, a12 x b21, a13 x b31, a21 x b11, a22 x b21, a23 x b31, a11 x b12, a12 x b22, a13 x b32, a21 x b12, a22 x b22, and a23 x b32)
- Is your theory correct? Does sugary drink consumption significantly mediate the relations between those causal antecedents and teeth condition? Does fruit consumption significantly mediate the relations between those causal antecedents and teeth condition? Does vegetable consumption significantly mediate the relations

between those causal antecedents and teeth condition? Which path is more strongly associated with teeth condition?

- Is your theory correct? Does sugary drink consumption significantly mediate the relations between those causal antecedents and weight status? Does fruit consumption significantly mediate the relations between those causal antecedents and weight status? Does vegetable consumption significantly mediate the relations between those causal antecedents and weight status? Which path is more strongly associated with weight status?

## Assignment 5-2

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**), the income level of child's household (**povlev4_21**), and their physical activity (**PhysAct_21**) serve as causal antecedent variables, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Please run a conditional process analysis to find out the answer. (Let's assume that these seven variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 5_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - income level of child's household (**povlev4_21**) : Low values indicate better lower income.
  - physical activity (**PhysAct_21**) : High values indicate more physcial activities.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.
  - fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
  - vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
  - teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
  - weight status (**BMI4_21**): High values indicate overweight.
- Draw the diagram to represent the paralell multiple mediator model with multiple causal antecedents and consequent variables.
- Write down the bmatrix.

- Write down the cmatrix.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a11, a12, a13, a21, a22, a23, a31, a32, a33, b11, b21, b31, b12, b22, b32, a11 x b11, a12 x b21, a13 x b31, a21 x b11, a22 x b21, a23 x b31, a31 x b11, a32 x b21, a33 x b31, a11 x b12, a12 x b22, a13 x b32, a21 x b12, a22 x b22, a23 x b32, a31 x b12, a32 x b22, and a33 x b32
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a11, a12, a13, a21, a22, a23, a31, a32, a33, b11, b21, b31, b12, b22, b32, a11 x b11, a12 x b21, a13 x b31, a21 x b11, a22 x b21, a23 x b31, a31 x b11, a32 x b21, a33 x b31, a11 x b12, a12 x b22, a13 x b32, a21 x b12, a22 x b22, a23 x b32, a31 x b12, a32 x b22, and a33 x b32)
- Is your theory correct? Does sugary drink consumption significantly mediate the relations between those causal antecedents and teeth condition? Does fruit consumption significantly mediate the relations between those causal antecedents and teeth condition? Does vegetable consumption significantly mediate the relations between those causal antecedents and teeth condition? Which path is more strongly associated with teeth condition?
- Is your theory correct? Does sugary drink consumption significantly mediate the relations between those causal antecedents and weight status? Does fruit consumption significantly mediate the relations between those causal antecedents and weight status? Does vegetable consumption significantly mediate the relations between those causal antecedents and weight status? Which path is more strongly associated with weight status?

## Assignment: Find an empirical journal article

- Find an empirical article in your field that uses the parallel multiple mediator model with multiple causal antecedents and consequent variables.
- What is the hypothesized model? That is, what is the diagram that visualizes the relations?
- Define Xs Ms, and Ys using layperson's terms

- Find out the values for all the path coefficients and indirect effects.
- Find out $R^2$
- Share your interpretation of the results for each path and indirect effect using layperson's terms
- Are the mediation effects significant? How do you know?
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 5 mins.
- **Submit your PPT slides to the designated folder on iStudy.**